

Host Restriction Mechanisms Linked to Enriched Motifs

CpG → ZAP / OAS3

ZAP binds CpG-enriched viral RNA; OAS3/RNaseL provides complementary restriction (independent but functionally overlapping pathways).

No CpG-enriched motifs in top 10

UpA → ZAP / OAS3

ZAP also binds UpA-enriched RNA; OAS3/RNaseL cleaves at UpA/UU sites. Mechanism of attenuation remains uncharacterized.

6/10 motifs contain UpA

Composition Bias

Purine-rich, GC-rich, AU-rich patterns may affect RNA structure and stability.

3/10 motifs purine/GC-rich